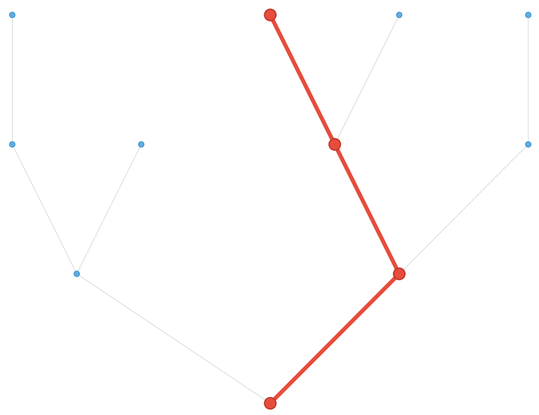
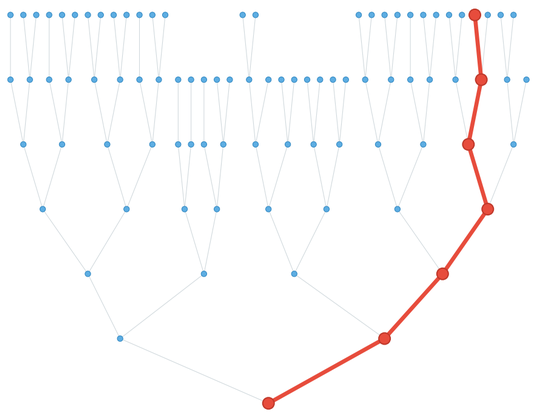
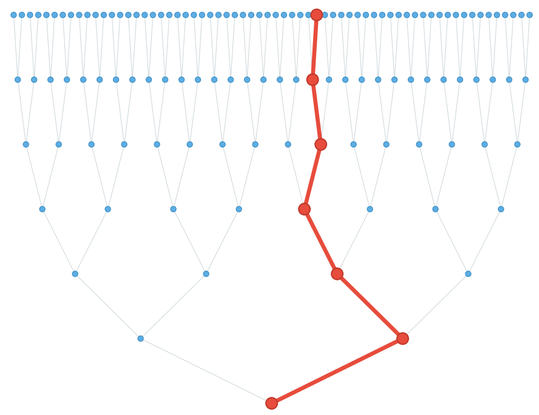
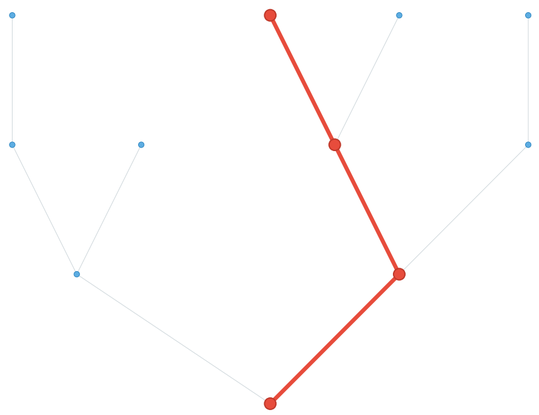
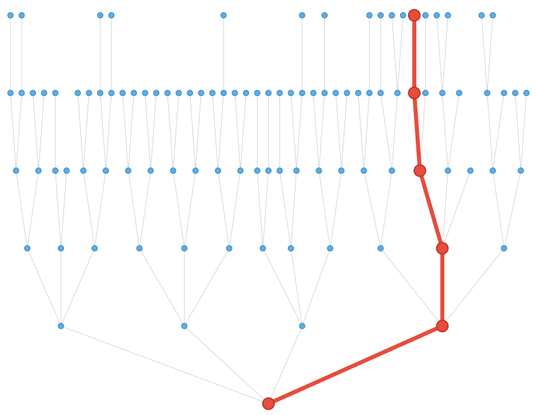
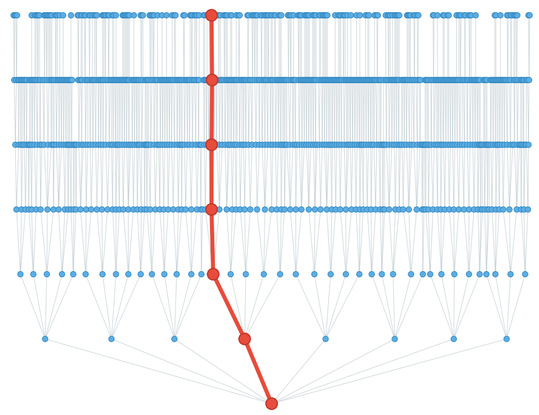
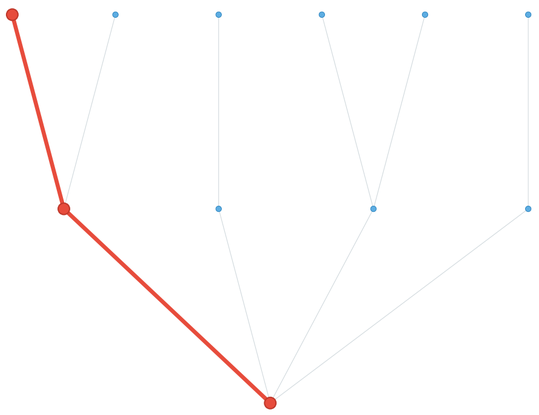
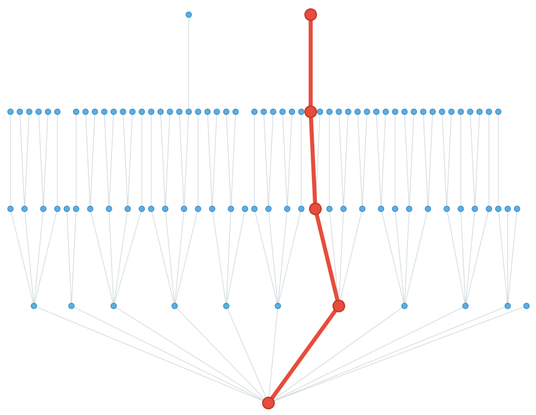
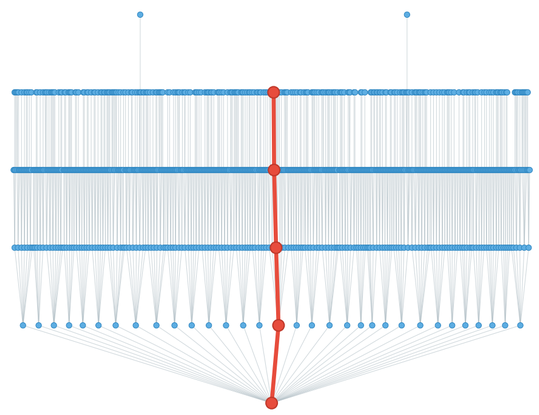
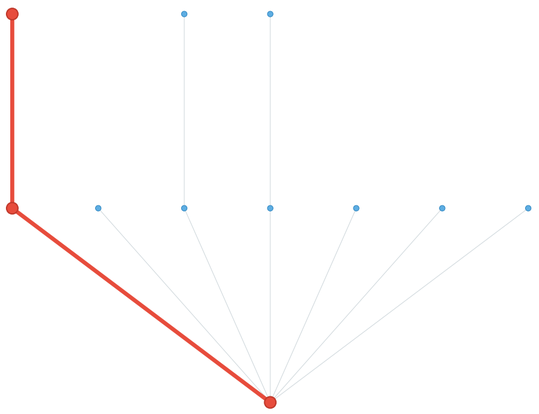
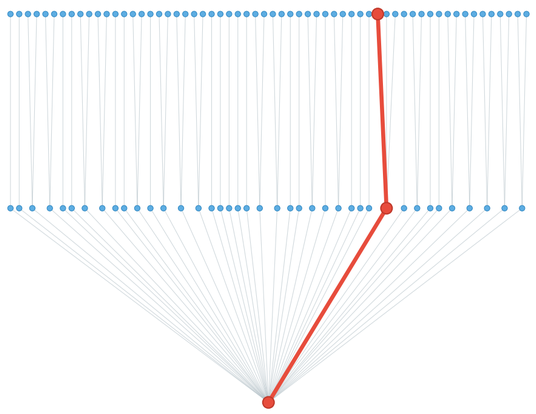
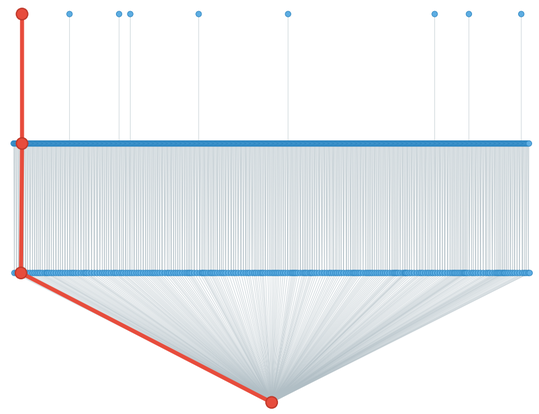
Effect of α_dpw on MCTS-DPW Tree Growth

Rule: if N^α ≥ num_children → sample new action (explore), else → UCB select (exploit)
Low α → narrow & deep (exploit-heavy) | High α → wide & shallow (explore-heavy)

	Iteration 1	Iteration 10	Iteration 100	Iteration 1000
α = 0.1	nodes=2 depth=1 root_children=1 max_width=1	 nodes=11 depth=3 root_children=2 max_width=4	 nodes=88 depth=6 root_children=2 max_width=29	 nodes=127 depth=6 root_children=2 max_width=64
α = 0.3	nodes=2 depth=1 root_children=1 max_width=1	 nodes=11 depth=3 root_children=2 max_width=4	 nodes=101 depth=5 root_children=4 max_width=42	 nodes=995 depth=6 root_children=8 max_width=382
α = 0.5	nodes=2 depth=1 root_children=1 max_width=1	 nodes=11 depth=2 root_children=4 max_width=6	 nodes=101 depth=4 root_children=11 max_width=51	 nodes=1001 depth=5 root_children=32 max_width=498
α = 0.8	nodes=2 depth=1 root_children=1 max_width=1	 nodes=11 depth=2 root_children=7 max_width=7	 nodes=101 depth=2 root_children=40 max_width=60	 nodes=1001 depth=3 root_children=252 max_width=739

Best path (most visited) | Other nodes | Best-path edges | Other edges